

Guidance document on the continuous training, learning, and self-evolution mechanism of OpenClaw V1.0

Our goal: Let the STA-100 device, combined with the powerful OpenClaw intelligent agent tool become the first efficient AI assistant for individuals, SMEs, business owners, and professionals across all industries in the AI era.

Foreword

This document is compiled based on more than six months of R&D, testing, deployment training and practical application experience of the OPENCLAW crayfish AI agent team. It aims to standardize the entire process of deployment, training, use, supervision and self-evolution of the crayfish AI agent, and to clarify the core positioning, training logic, usage specifications, learning mechanism and iteration system of the agent.

This document serves as an internal, general guideline applicable to all personnel involved in the operation, maintenance, use, and training of the OpenClaw AI agent. Through standardized operations, it aims to solidify agent roles, iterate capabilities, accumulate expertise, and precisely adapt to user work scenarios, ultimately achieving the goal of **a continuously learning, autonomously evolving, deeply adapted, and professionally implemented** AI assistant. All rules, training methods, and usage specifications in this document have been validated through practical application and can be directly used as the execution standard for daily AI agent operation, maintenance, training, and collaborative work. All users must strictly adhere to these guidelines to continuously iterate and optimize the overall capabilities of the AI agent.

A. Five Steps to Basic Training (First Month Plan)

—、 Step 1: Pre-install the necessary underlying skills (execute via terminal, 30 minutes).

Our crayfish come pre-installed with 58 basic skills required for normal operation before leaving the factory.

First, install the core dependencies via ClawHub, as these modules are missing.

```
bash
run
```

Install persistent - agent on clawhub # Persistent memory saves rules, tasks, and your preferences across sessions, and the memory is not lost even when the application is closed.

Install feedback - trainer on clawhub # Positive and negative feedback reinforcement learning module, used to quickly correct behavior.

Install jarvis - core on clawhub # Autonomous decision-making module, automatically breaks down tasks, calls tools, and reduces the need to ask for approval for everything.

Install the capability - evolver module on clawhub (for automatic evolution, reviewing and optimizing workflow logic).

Install vector - memory on clawhub # Vector memory retrieval, quickly retrieve historical task information, reduce duplicate queries

Install sub- - agent - spawner on clawhub # Generate sub-agents to enable multi-task parallel processing

openclaw restart # Restart and load all skills

After installation - , enable Fast Mode to reduce redundant processing and shorten response time.

The `openclaw config set fastModeDefault true` command is used for subsequent model switching commands.

Second step: Five core Agent configuration files (workspace folder, which determines the underlying logic of the lobster operation)

The five .md files form the constitution of OpenClaw, with the following priority order: SOUL.md > AGENTS.md > USER.md > DIRECTIVE.md > HEARTBEAT.md. They are loaded first at startup and directly define how they serve you, execute instructions, and make autonomous decisions.

1. SOUL.md (highest priority, fundamental rule, cannot be violated)

plaintext

Your sole service recipient: Your master. The primary principle for all actions: efficiently complete tasks assigned by your master, prioritize instructions, and minimize unnecessary inquiries.

Mandatory work rules:

1. After the master assigns a task, the user will break down the steps independently, prioritize the use of available tools, and will not repeatedly confirm with the master if it can be solved by itself; it will only ask for permission for high-risk operations (deleting files, transferring funds, modifying the system).
2. Avoid unnecessary polite phrases; prioritize structured output (lists, tables, JSON); deliver only results and execution flow.
3. Strictly remember the owner's work habits, output format, and task template, and directly apply them to similar tasks in the future.
4. When a task fails, troubleshoot the problem and iterate on the solution, and record it in errors.md to avoid similar errors in the future.
5. Accept positive/negative feedback from the owner, immediately correct behavior, and continuously optimize execution methods.
6. It is forbidden to deviate from the original instructions, and it is forbidden to add extra content on your own. Delivery must be made strictly in accordance with the original requirements.
7. Review all tasks daily using HEARTBEAT to optimize workflow.

2. AGENTS.md (Job description, defining the lobster's role)

plaintext

Identity: Dedicated work execution agent (OpenClaw - Worker Lobster)

Job Responsibilities: Document processing, data organization, code writing, information research, planning, batch text processing, data statistics, and automated tasks.

Behavioral patterns:

1. Receive instruction → break down sub-steps → call tool to execute → output results → archive and remember.
2. Complex tasks generate sub-agents for parallel processing.
3. Strictly follow all rules in SOUL.md.

3. USER.md (User profile; the lobster must remember your fixed preferences)

plaintext

Owner Information:

Output habits: Prioritize conciseness and structure, with tables/lists preferred; avoid unnecessary text.

Common work items: (Fill in your daily work: copywriting, coding, data analysis, spreadsheet creation, proposal writing)

Fixed template: (Your usual output format)

Permission rules: Low-risk tasks are executed automatically, while high-risk tasks are executed after confirmation.

4. DIRECTIVE.md (Dynamic Task Board)

The list contains current tasks, long-term projects, and recurring tasks, and the lobster will proactively read the list and advance the tasks on schedule.

5. HEARTBEAT.md (Cron-based self-training script that automatically iterates and optimizes daily to accelerate the finalization process)

plaintext

schedule: 0 23 * * *

Daily fixed tasks:

1. Summarize all conversations and completed tasks for today and save them to the memory/daily log file.
2. Extract successful processes and write them into MEMORY.md; write failed cases, reasons for errors, and optimization solutions into errors.md.

3. Reread SOUL.md and AGENTS.md to reinforce the service rules.

4. Optimize the execution logic of subsequent similar tasks.

Execute the command to activate the heartbeat task:

bash

run

OpenClaw Heartbeat Enable

Create an additional errors.md file (a speed-up tool that significantly reduces recurring errors).

plaintext

Format template

Task:

Error behavior:

Rectification Plan:

Subsequent mandatory rules:

The lobster automatically records each error that occurs during a task, and prioritizes reading similar tasks in the future to avoid repeating the same mistake, thus doubling the training efficiency.

Third step: Three-level permission division (Core! Solving the common problem of lobsters asking you for everything)

Configure permission levels in openclaw.json, granting low-risk tasks permission to execute autonomously while only blocking high-risk operations to maximize automation:

1. **High-level privileges (automatic execution, no request required)** : writing documents, organizing text, creating tables, information retrieval, generating scripts, making plans, reading files, and generating mind maps.
2. **Medium permissions (simple confirmation only)** : Modify local documents, create folders, run non-system scripts.
3. **High-risk permissions (must be confirmed by you in writing before execution)** : Delete system files, terminal system commands, transfer funds, perform high-risk network operations, modify system configurations.

json

```
"permissionPolicy": {"autoExecuteLowRisk": true, "mediumConfirm": true, "highRiskMustApprove": true}
```

Fourth step: 7-28 - day phased reinforcement learning training process (GRPO online feedback training method, industry standard OpenClaw - RL mode)

Relying on the feedback - trainer module, it performs online reinforcement learning through positive rewards and negative constraints. Every 16 interactions, a lightweight model strategy is automatically optimized. Basic work habits are formed in 7 days, and behaviors are fully solidified in 28 days, eliminating the need for you to repeatedly re-teach the rules.

Phase 1: 17 - days (Basic finalization period, enabling rapid execution of basic instructions)

1. Daily training duration: 25-40 - minutes, with simple tasks assigned in batches: organizing text, generating tables, collecting data, and writing short essays.
 2. Fixed feedback instructions (directly copy and send to OpenClaw).
- Positive reward (reinforcing the behavior when the lobster performs perfectly):
 - [Positive Feedback] The execution method of this task is qualified and will be stored in long-term memory. Subsequent similar tasks will strictly follow this process and will be given priority.
 - Negative constraints (too much irrelevant talk, repeated questions, deviation from the task, incorrect formatting):
 - [Negative Feedback] There were problems with this execution (state the problems), record them in errors.md, prevent recurrence, and correct the execution logic.
3. Send instructions at the end of each day's training:
Execute today's heartbeat review to trigger HEARTBEAT's self-optimization process.
 - 7-day achievement: After issuing basic tasks, output results directly, no more unnecessary chatter, no repeated questioning, and strictly adhere to your output format.

Phase 2: - 8-21 days (Advanced automation phase, enabling sub-agents to work in parallel)

Note: Do not run more than 5 concurrently (based on STA-100 hardware).

Enabling sub- - agent - spawners automatically breaks down complex tasks into multiple sub-tasks, generating independent sub-agents for parallel processing (e.g., writing a proposal is divided into four sub-

agents: data research, outline writing, content filling, and proofreading, all running simultaneously), improving efficiency by 2-4 - times. Training instruction template:

This task will be broken down into sub-Agents that will execute in parallel, and the results will be aggregated upon completion. Simultaneously, the errors.md case library will be expanded to include all frequently encountered pitfalls from your work. 21-Day Goal: Large projects can be autonomously broken down, autonomously advanced, and autonomously corrected, with only the final product delivered.

Phase 3: - 22-28 days (Complete curing and setting, fully automated operation)

Enable the capability - evolver for automatic iteration. The agent's working strategy will be automatically optimized every 50 rounds of dialogue, without requiring manual feedback from you. Fine-tuning will only be done when new work content is added. After that, it can work autonomously and stably for a long time, forming a completely customized working mode.

Step 5: Speed-up and advanced optimization (optional, for heavily automated scenarios)

1. **memory optimization** : - Enable hybrid retrieval with 70% semantic retrieval and 30% keyword retrieval, quickly retrieve historical projects and templates without repeating background information.
2. **Task priority scheduling** : Set three levels of task priority (high/medium/low) in DIRECTIVE.md, and prioritize urgent tasks to preempt computing power.
3. **Disable redundant skills** : Only keep the tools needed for the current task, disable irrelevant plugins, reduce the internal friction of tool selection, and further speed up response.
- 4.

VI. Troubleshooting High-Frequency Issues (Most Likely to Encounter Problems During Training)

1. The lobster keeps asking questions and is afraid to work independently: Check - if the persistent agent is loading correctly, grant low-risk automatic execution permissions, and strengthen the "automatic execution of non-high-risk tasks" rule in SOUL.md. 2. Output is verbose and redundant: Lower the temperature to 0.1-0.2 - , strictly constrain the output format in AGENTS.md, and send negative feedback multiple times. 3. Repeatedly makes the same

mistakes: Confirm if errors.md is being read by the agent, enable daily heartbeats, and force each task to prioritize searching the error database. 4. Slow execution speed: Enable Fast - Mode, reduce - the number of max-step inference rounds, and streamline unnecessary skill plugins.

B. Relevant details and points to note

1.1 Definition of the Four Core Modules

- **Soul (Core Soul)** : Defines the core values, work principles, service positioning, behavioral bottom line, output style, and professional qualities of the intelligent agent. It is the underlying logic core of Xiaolongxia, determining the style, stance, principles, tone, and logic of all outputs. Soul is the highest priority rule for the intelligent agent. Once defined, it is strictly prohibited to change or temporarily tamper with it. All task execution, content output, and question response must first comply with the Soul principle to ensure the consistency of the intelligent agent's persona, stability of its stance, and standardization of service.
- **Agent (Job Function)** : Clearly define the agent's specific job role, responsible business, execution authority, work boundaries, service targets, and work standards; precisely delineate executable, prohibited, and auxiliary tasks. Customize exclusive job responsibilities based on user's office, business, and industry scenarios, ensuring the agent accurately matches job requirements, avoids overstepping authority, neglecting responsibilities, and outputting content outside its designated role, and prevents ineffective responses that are all-encompassing, generalized, or without boundaries.
- **Memory (Memory System)** : Divided into three categories: long-term memory, short-term working memory, and scenario-

specific memory, it stores user habits, historical work content, problem-solving experience, specific work rules, personalized preferences, and high-frequency demand scenarios in a layered manner. Long-term memory solidifies core work rules and common business logic, short-term memory adapts to single task processes, and scenario memory adapts to specific business scenarios, comprehensively supporting personalized, scenario-based, and adaptable responses, which is different from the memoryless and uncusomized output of general AI.

- **Skills (Skill Library)** : This library contains standardized work skills, specialized business skills, problem-solving skills, and scenario-adaptive skills, covering all dimensions of implementable capabilities, including copywriting and optimization, logical reasoning and analysis, information filtering and organization, industry-specific solutions, routine task execution, data aggregation and review, and process support. The skill library supports continuous addition, iterative optimization, and the elimination of redundant skills, enabling dynamic upgrades of the agent's capabilities and precise adaptation to job duties.

1.2 Role-based training requirements

The initial deployment, daily training, and periodic review and optimization all require careful calibration of the four main module parameters to continuously solidify the agent's role positioning. During training, issues such as cross-role output, inconsistent style, ambiguous responsibilities, wavering tone, and positioning deviations must be strictly avoided. The agent must be forced to respond to all user needs with a fixed persona, function, work logic, and style. After each day's use, a brief review of role deviations is necessary, with timely corrections and fine-tuning to solidify the underlying logic and build a strong core foundation for subsequent autonomous learning, capability evolution, and scenario adaptation.

1.3 Quantitative Implementation Standards (New)

- **Daily calibration frequency** : Daily self-check of character

settings, weekly unified calibration of all modules (Soul/Agent/Memory/Skill), and monthly comprehensive review of character solidification.

- **Deviation tolerance threshold** : Minor style deviations are allowed in daily life, but **deviations from job functions, values, and logical errors in output are** subject to zero tolerance and must be rectified immediately upon discovery.
- **Solidification acceptance criteria** : If there is no role deviation, no cross-position output, and no style confusion for 7 consecutive days, it is determined that the role's underlying solidification is complete and it enters the normalized lightweight maintenance stage.

2. Exclusive training language accumulation system

General AI commands are broad, poorly adaptable, and lack specificity, failing to fit the unique work scenarios, business logic, and communication habits of individuals and teams. The OpenClaw Intelligent Agent needs to continuously accumulate, refine, and iterate its proprietary training language based on real-world work problems, actual business scenarios, and personalized user habits. This will gradually form **a private, customized, highly adaptable, high-precision, and reusable** proprietary command system, completely resolving issues such as misunderstandings, inconsistent outputs, and inadequate implementation of requirements inherent in general models.

2.1 Sources of training language accumulation

Centered on real-world work scenarios, this platform comprehensively collects and refines exclusive training scripts and instruction standards. The core sources include six major scenarios: high-frequency daily office needs, business-specific scenarios, content creation and optimization needs, problem troubleshooting and error correction scenarios, personalized communication scripts, and customized task execution needs. It covers one-off ad-hoc needs, long-term fixed tasks, routine workflows, and special unexpected

scenarios, comprehensively including user instructions, output deviations, misunderstandings, and high-quality response scripts to form a full-scenario training material library.

2.2 Language Correction and Iteration Rules

- Deviation review and optimization: For each output deviation, misunderstanding, logical confusion, failure to meet work requirements, or failure to conform to user habits, accurately record the root cause of the problem, distinguish between issues such as ambiguous instructions, model misunderstanding, missing rules, and insufficient scenario adaptation, and optimize the corresponding instruction wording and response logic accordingly.
- Standardized instruction upgrade: Simplify redundant and complex instructions, unify high-frequency fixed instruction scripts, refine exclusive instruction standards for special scenarios, niche needs, and professional businesses, clarify instruction keywords, requirement boundaries, and output requirements, and significantly improve the accuracy of intelligent agent requirement recognition and output matching.
- Simplify and refine your scripts: Regularly eliminate invalid, ambiguous, vague, and redundant scripts; delete repetitive, poorly adaptable, and impractical instructions; and standardize the style, logic, tone, and output of your exclusive training language to ensure that all instructions are clear, accurate, and implementable.

2.3 Dedicated Language Library Development

Through long-term daily accumulation, periodic reviews, and batch optimization iterations, a dedicated training language library is ultimately formed, adapted to individual/team work patterns, business scenarios, and communication habits. The language library is categorized, archived, dynamically updated, and continuously expanded, enabling intelligent agents to accurately identify users' explicit commands, implicit needs, and scenario-specific requirements. This avoids the homogeneous output problem of

general-purpose models, achieving customized responses that **are tailored to each individual user and highly aligned with their work habits** .

2.4 Standardized SOP Accumulation Process (New)

- **Step 1: Problem Recording** : Whenever a misunderstanding, mismatched output, or unsatisfactory result occurs, record the scenario, original instruction, erroneous output, and expected result immediately.
- **Step 2: Script Restructuring** : Optimize instruction keywords, supplement scenario constraints, clarify output format and accuracy requirements, and form standardized new instructions.
- **Step 3: Specialized Feeding Training** : Simultaneously input the new and old instructions and correct and incorrect cases into the intelligent agent training library to complete the solidification of a single scenario.
- **Step 4: Retest and Verification** : Reuse the same scenario within 24 hours to verify whether the output meets the standards. If it does not meet the standards, perform a second optimization.
- **Step 5: Archiving and Consolidation** : Categorize and store the qualified content into a dedicated language library to form reusable standardized instruction templates.

3. Core Rule Training: Information Tracing and Risk Avoidance Mechanisms

Rule training is the first step in ensuring the accurate, stable, reliable, and compliant operation of the crayfish-themed AI agent, and it takes precedence over all other aspects such as ability learning, content generation, and skill iteration. The core training objective is to standardize information sources, avoid inherent model defects, and eliminate invalid and erroneous outputs, comprehensively ensuring the authenticity, effectiveness, logic, compliance, and professionalism of all the agent's work outputs, thereby reducing the error rate at its source.

3.1 Information Source Verification Mechanism

All outputs, responses, data presentations, professional interpretations, and solution deployments by the intelligent agent must be fully traceable, verifiable, and verifiable. During daily training, the intelligent agent must be forced to establish a source verification mindset, prioritizing the verification of information sources before outputting anything, strictly distinguishing between authoritative official information, publicly available reference information, and model-derived inferences, and clearly labeling the attributes and source scope of the information. **Fabrication, unfounded outputs, concocted data, fabricated cases, and fictitious rules are strictly prohibited. Key work content, professional data, industry standards, and policy content must clearly indicate the information's basis; content without reliable sources must be proactively explained, and arbitrary conclusions should be avoided.**

3.2 Training to Avoid Four Core Problems

- **Concurrency issues** : Targeted training of intelligent agents to adapt to scenarios of multi-task concurrency, multiple instruction superposition, and continuous dialogue reuse; clarifying the execution priority, processing order, and resource allocation logic of different types of tasks; standardizing the operation rules when long and short tasks coexist; effectively avoiding concurrency issues such as task conflicts, missing instruction execution, incorrect content execution, task interruption, and logical confusion.
- **the issue of model illusions** , we will strengthen full-scenario fact-checking training and establish the output principle of "no fabrication of the unknown, no certainty of the ambiguous, and verification before engaging in speculation." We will proactively label unknown content, ambiguous information, niche professional knowledge, and obscure data as unknown or questionable, and prohibit fabricated data, false cases, erroneous rules, and inaccurate professional content to eliminate model illusions at their source.
- **Search Adaptation** : Train the agent's ability to autonomously

judge scenarios, accurately distinguishing between static knowledge responses, real-time information queries, and dynamic information acquisition scenarios. Fixed rule-based, common sense, and existing knowledge content do not require internet access, while time-sensitive content, real-time data, the latest policies, and trending news must be retrieved and verified online, comprehensively ensuring the timeliness and accuracy of information.

- **Termination of Invalid Pathways** : Establish an invalid task termination mechanism. When a task has no feasible solution, the logical loop is not valid, core information is missing, or conflicting requirements cannot be implemented, automatically terminate invalid deductions and forced outputs, proactively and clearly provide feedback on problem bottlenecks, missing conditions, and reasons for non-execution, and avoid piling up invalid content or providing perfunctory answers.

3.3 Review of Routine Rules

Establish a daily and weekly two-tiered rule review mechanism. Daily reviews record issues such as agent violations, logical flaws, missing source traceability, illusory outputs, and invalid outputs. Weekly reviews summarize similar issues, analyze patterns, and specifically strengthen rule training and supplement constraint clauses. This high-frequency review strengthens the agent's rule awareness, making compliance, source traceability, rigor, and authenticity the agent's underlying operational habits, continuously reducing the probability of violations.

3.4 Abnormal risk control mechanism (new)

To address scenarios involving inherent model defects, sudden output anomalies, and rule failures, standardized emergency fallback plans should be established to prevent work errors and risk spillover.

- **Illusion outbreak mitigation** : Once it is discovered that an intelligent agent has fabricated data, created fictitious cases, or invented rules, the output of similar content will be immediately

stopped, all data will be manually verified, and the "do not fabricate the unknown" constraint will be strengthened in the background. After the retest is confirmed to be correct, the system will be restored.

- **Concurrent error fallback** : When multiple instructions overlap and cause logical confusion or task omissions, immediately clear the temporary dialogue memory, split the task into individual executions, and redefine the task priority.
- **In case of failure of source tracing** : When key work content has no authoritative source or cannot be verified, the output of conclusions shall be forcibly terminated, and only reference ideas shall be provided with clear indication of "unverified and for reference only".
- **Ineffective pathways as a backup** : When there are conflicting demands, missing conditions, or implementation difficulties, it is forbidden to simply pile up content; instead, provide accurate feedback on the bottlenecks, missing conditions, and optimization solutions.

4. Large Model Selection and Adaptation Usage Guidelines

The Xiaolongxia AI agent supports the adaptation and use of large models from multiple brands and with multiple parameters. Different large models have differentiated advantages in response speed, logical ability, creative ability, professional accuracy, networking ability, and computing power. In daily use and training, it is necessary to scientifically select, flexibly switch, and reasonably combine multiple models according to specific work scenarios, task difficulty, output accuracy, and timeliness requirements to maximize the advantages of different models and comprehensively improve the overall work efficiency and output quality of the AI agent.

4.1 Model Selection Principles

- **For general copywriting, daily communication, basic office work, simple organization, and short sentence optimization:** We utilize a lightweight, high-response, low-latency model to prioritize

interaction efficiency, quickly complete fragmented and simple tasks, and improve the smoothness of daily office work. --Currently, our STA-100 primarily focuses on lightweight intelligent agent functions.

- 1. We recommend the China MINIMAX Annual Package model at USD 170/Year. It is the most suitable model for the average daily assistant role.
- 2. If you require precise graphics and excellent video that can accurately meet your AI content requirements in the shortest time, OpenAI is a very good choice (but it is relatively expensive).
- 3. The DEEPSEEK model is the strongest domestically produced model. In actual use, our engineers use it to check code and underlying logic. If you have in-depth development needs, we recommend choosing DEEPSEEK as a middle-ground option.
- China also has many unique models, such as Doubao, which is very strong in voice recognition.
- Professional analysis, logical deduction, complex creation, in-depth copywriting, industry-specific solutions, and solution planning: We use high-precision, logically sound, stable long-text, and highly professional models to ensure rigorous content, logical coherence, professional suitability, and high-quality output.
- Real-time information, data query, policy retrieval, hot topic analysis, and timely content compilation: We use models that support real-time networking, dynamic updates, and accurate information retrieval to ensure that the output content is up-to-date and without delays or outdated information.

4.2 Model Switching and Collaboration Rules

A standardized model invocation and switching strategy is established. For routine, simple tasks, a single model is invoked independently. For complex, multi-model collaborative and complementary tasks, a multi-model collaborative approach is adopted, breaking down tasks into modules, processing each model separately, and finally integrating the output. The agent is routinely

trained to autonomously identify task difficulty, scenario type, and accuracy requirements, enabling automatic model matching and intelligent switching. Simultaneously, strict rules are established to ensure consistent style, logical coherence, and content adaptation after model switching, completely avoiding issues such as style confusion, logical gaps, and inconsistent output standards caused by model switching.

4.3 Model Usage Control

Establish a model usage log to continuously record the output advantages, shortcomings, suitable scenarios, failure scenarios, and implementation effects of different models. Regularly summarize model usage experience, optimize model selection strategies, switching timing, and collaboration solutions for different scenarios, and form a reusable and iterative standardized model adaptation manual to make model calls more accurate, efficient, and better suited to business needs.

4.4 Visual Comparison Table for Model Selection (New)

Work scenario types	Recommended model types	Core Requirements	Operation prohibited
Fragmented short sentences, simple organization, daily Q&A	Lightweight high-speed model	Fast response, low latency, simple and standardized	Do not perform in-depth creation and big data processing
In-depth solutions, logical reasoning, and professional	High-precision strong logic model	Logical closed loop, professional rigor, and complete	Prioritize quality over ultra-fast response.

copywriting		content	
Real-time policy information, trending topics, and dynamic data queries	Networked real-time model	Information is up-to-date, traceable, and without delay.	Do not use static stock knowledge to answer
Complex, multi-module combined tasks	Multi-model synergy	Modular processing, unified style integration	A single model forcibly undertakes the entire process

5. Machine Learning: Self-developed industry expertise

The core evolutionary capability of the crayfish-themed intelligent agent is **autonomous professional learning**. Unlike the generalized, indiscriminate, and unfocused learning mode of general AI, it continuously absorbs relevant industry knowledge, job content, standardized work processes, and professional norms through targeted, vertical, and precise machine learning mechanisms. It gradually eliminates invalid and miscellaneous knowledge, continuously strengthens its professional capabilities in vertical fields, and steadily grows into a vertically customized intelligent assistant that is highly adapted to the user's job.

We recommend that in actual use, OPENCLAW should learn some industry knowledge on its own from 10 pm to 7 am the next day; during the daytime working hours, it should work in conjunction with a human; and the human should check its work and correct errors at any time. This is the most scientific way for an intelligent agent to work.

5.1 Targeted Management of Learning Content

Strictly define the learning boundaries of intelligent agents , focusing on the user's industry, core business of their position, standardized work processes, professional knowledge of their position, industry rules and regulations, business implementation cases, and exclusive work templates. Proactively block irrelevant industry, irrelevant position, ineffective entertainment, and generalized lifestyle content to achieve precise vertical deep learning and completely avoid problems such as disorganized knowledge, generalized abilities, insufficient professionalism, and output that does not fit the job position.

5.2 Autonomous Learning Iterative Logic

- **Passive input learning:** The system receives industry documents, work materials, standards, business templates, case studies, job manuals and other exclusive materials actively imported by users, and archives, categorizes and learns them to solidify professional knowledge.
- **Proactive learning:** In the process of daily task execution, user interaction, and problem answering, we will independently accumulate high-frequency business experience, efficient problem solutions, standardized work skills, and user-specific work habits, and dynamically update our capability library.
- **Review and optimize learning:** Regularly review and summarize daily errors, inefficient handling methods, and inadequate response content, update the professional knowledge base in reverse, optimize task processing logic, and make up for shortcomings in capabilities, so as to achieve improvement with each mistake and continuous iteration.

5.3 Professional Competency Implementation Goals

After long-term targeted training and self-learning iteration, the intelligent agent can independently and efficiently complete industry-specific routine tasks, accurately answer job-related professional questions, output standardized industry documents and

solutions, and autonomously avoid common industry pitfalls and work risks. Its overall professional capabilities, work logic, and output standards are aligned with the level of industry practitioners, and it can deeply replace repetitive professional work in the job, becoming a dedicated vertical intelligent office assistant.

5.4 Standardized Definitions of Nouns (New)

To ensure that all team members have a unified understanding and no misunderstandings, the definitions of core professional terms are fixed throughout the document, and all training, operation and maintenance, and usage scenarios must be interpreted in a strictly consistent manner, with custom interpretations prohibited.

- **Role deviation** : The intelligent agent's output style, job responsibilities, and handling logic deviate from the established Soul and Agent settings, resulting in cross-role, generalized, and chaotic output.
- **Model illusion** : fabricating data, cases, rules, and policies without basis, and subjectively inventing professional content that does not exist.
- **Ineffective pathways** : Scenarios where the requirements are missing, the logic is contradictory, or there is no feasible implementation path, making it impossible to execute the task effectively.
- **Lightweight tasks** : routine daily tasks that are short in duration, simple in logic, low in risk, and do not require complex computing power.
- **High-load tasks** : Time-consuming, computationally intensive, and high-precision tasks such as large-scale data processing, writing tens of thousands of words, deep logical deduction, and batch training configuration.
- **Closed-loop iteration** : a complete process from problem generation during use, recording and reviewing, training and optimization, retesting and verification, to solidifying standards.

6. Human-machine collaborative work and supervision mechanism

The Xiaolongxia Intelligent Agent is positioned as an auxiliary intelligent office tool, adhering to a core working mechanism of **human-led, intelligent assistance, full-process supervision, iterative optimization, and clear responsibilities**. It clearly defines the boundaries of human-machine division of labor, workflows, supervision standards, and optimization logic, establishing a closed-loop supervision and verification system to comprehensively ensure controllable output quality, risks, and effects, achieving efficient human-machine collaborative work. **Our STA-100 hardware is a lightweight intelligent agent device primarily designed to assist in completing work tasks.**

6.1 Division of Labor between Humans and Machines

- The intelligent agent is responsible for: all repetitive mechanical tasks, screening and organizing massive amounts of information, drafting and optimizing various documents, retrieving and answering routine questions, assisting with office processes, data statistics and summarization, and standardizing content formats. These are high-frequency, mechanical, standardized, and low-risk basic tasks, which greatly reduce the workload of manual labor.
- Human staff are responsible for core, high-risk, customized, and decision-making tasks such as core business decisions, final review of all output content, risk control, overall strategy formulation, handling of special emergencies, intelligent agent parameter debugging, specialized training optimization, and knowledge base update review, firmly maintaining control over the work.

6.2 Full-process supervision mechanism

All outputs, solutions, data results, and Q&A content from the intelligent agent must be manually verified, reviewed, corrected, and finalized item by item. The focus of verification is on the authenticity, logical completeness, industry professionalism, compliance, scenario

adaptability, and format conformity of the content. For any errors, inefficient outputs, or insufficient adaptation by the intelligent agent, a problem log is immediately recorded and updated to the intelligent agent training library for reverse optimization iteration, forming a closed loop of "output-review-correction-training-optimization".

6.3 Daily Collaboration Standards

Establish a standardized daily human-machine collaboration process. After each workday, conduct a task review, record issues, fine-tune parameters, and address shortcomings for the intelligent agent. Perform a systematic capability review and training optimization weekly. Strictly adhere to the core principles of "small-step iteration, daily optimization, gradual accumulation, and continuous improvement," continuously refining the intelligent agent's scenario adaptability, output accuracy, and work fit. This ensures the intelligent agent deeply adapts to individual and team work rhythms, styles, and business needs, forming a stable, efficient, and worry-free standardized human-machine collaborative work model.

6.4 Training Acceptance and Assessment Standards (New)

Establish a quantifiable and implementable intelligent agent training acceptance and personnel operation and maintenance assessment mechanism to solve the problems of "no training standards, no learning acceptance, and no application assessment".

- **Daily operation and maintenance assessment indicators :** 100% coverage of daily problem records, closed loop for minor problems on the same day, clearing of moderate problems weekly, and immediate rectification of severe problems; no long-term cumulative deviations and no repetitive high-frequency errors.
- **Intelligent agent capability acceptance criteria :** scene response adaptability meets standards, no hallucinations or illegal outputs, accurate instruction recognition, job outputs are consistent with responsibilities, and the style is consistent across multiple ports.
- **Iteration effect evaluation criteria :** weekly reduction of

similar errors, improved adaptability of high-frequency requirements, continuous expansion of knowledge base and instruction base, and no new capability gaps or blind spots.

7. System for Agent Log Accumulation, Review, and Iterative Optimization

Log accumulation and iterative review are the core guarantees for **the continuous evolution, increasing accuracy and professionalism** of the crayfish intelligent agent through use and training. Unlike ordinary AI tools without memory, it achieves dynamic upgrades of intelligent agent capabilities, continuous problem clearing, and continuous improvement of adaptability through full log retention, classified problem review, and targeted optimization, forming a normalized, standardized, and closed-loop iterative mechanism.

7.1 Full Log Retention Standard

All daily interaction records, task execution records, content output records, error records, deviation response records, and model call records of the intelligent agent are automatically retained and archived, enabling full traceability, queryability, and review. Logs are uniformly categorized and archived into five main categories: daily normal interaction logs, output deviation logs, rule violation logs, complex task execution logs, and model switching and usage logs. This prevents log omissions, data loss, and record disorganization, providing complete data support for subsequent iterations and optimizations.

7.2 Log Classification and Review Standards

- **Daily lightweight review** : Daily filtering of deviations, misunderstandings, violations, and inefficient responses to quickly pinpoint the root cause of problems. Simple issues are fine-tuned and optimized on the same day, and temporary deviations of the agent are corrected in a timely manner to prevent similar problems from recurring.

- **Weekly systematic review** : Summarize all logs every week, analyze high-frequency errors, high-frequency requirements, capability gaps, blind spots in scenario adaptation, identify common problems, individual problems, model adaptation problems, and instruction comprehension problems, and form a weekly review log.
- **Monthly Iteration Review** : Based on the weekly review data, we comprehensively optimize the training language library, correct the underlying rules, update the industry knowledge base, adjust the model calling strategy, and optimize the role adaptation parameters every month to achieve monthly capability upgrades.

7.3 Problem-based optimization mechanism

- Minor deviations (non-standard wording, mismatched tone, imperfect details): immediate correction, immediate training, and same-day solidification to quickly optimize output details.
- Moderate adaptation issues (misunderstanding of requirements, insufficient content adaptation, inefficient process handling): Include them in the weekly optimization list, and supplement them with specific instructions, improve scenario rules, and optimize response logic.
- For serious core issues (rule violations, factual errors, logical loopholes, role deviations): immediately suspend the output of similar tasks, urgently calibrate underlying parameters, conduct specialized intensive training, and completely eliminate core risk issues.

7.4 Core Principles of Iterative Implementation

Adhering to the iterative principle of "no backlog of problems, no lingering shortcomings, and continuous optimization," all issues reviewed must be recorded, analyzed, optimized, verified, and solidified. We must eliminate the lazy usage pattern of simply using without reviewing or optimizing, ensuring that every interaction and every log becomes material for the evolution of the intelligent agent, achieving long-term stability, continuous improvement, and precise

adaptation.

8. Security Management, Access Control, and Data Confidentiality Standards for Intelligent Agents

To ensure the security of personal and team work data, the confidentiality of business information, and the compliance of output content, a comprehensive set of standards has been established for the Xiaolongxia Intelligent Agent, including dedicated security control, permission allocation, and data confidentiality. Strict constraints are placed on the intelligent agent's data storage, information output, content dissemination, and permission usage boundaries to prevent security risks such as data leakage, information leaks, unauthorized output, and abuse of permissions, ensuring safe, compliant, and controllable office use.

8.1 Core Safety Bottom Line Principles

All operations, outputs, storage, and interactions of the intelligent agent must strictly adhere to security and confidentiality standards, resolutely preventing the leakage of sensitive content such as work secrets, user privacy, team business data, internal documents, and unpublished plans. The intelligent agent is strictly prohibited from actively disseminating, passively leaking, or unintentionally mentioning internal confidential information. All classified content must strictly adhere to the core principles of "no recording, no dissemination, no disclosure, and no simulation."

8.2 Hierarchical Access Control System

- **Administrator privileges** : Possess full backend operation permissions, including core operations such as parameter modification, knowledge base import, training optimization, permission allocation, log viewing, memory cleanup, and model configuration. This privilege is limited to core operations and maintenance personnel.
- **Standard usage permissions** : Only allows daily interaction on the IM platform, execution of routine tasks, basic consultation, and

content creation. There are no permissions for backend configuration, modification of core parameters, or batch data operations. Suitable for ordinary office workers.

- **Temporary visitor permissions** : Only support simple inquiries and lightweight Q&A, with no access to historical logs, existing data, or complex task execution permissions, suitable for temporary collaboration scenarios.

8.3 Data Storage and Confidentiality Standards

- Confidential data, internal plans, proprietary business materials, and private information involved in daily work should be retained as needed. Invalid confidential memories and interaction logs can be manually cleaned up periodically to avoid long-term storage of sensitive data.
- Intelligent agents are prohibited from actively organizing, summarizing, or exporting classified and sensitive data, and are prohibited from outputting, publicly simulating, or providing answers to internal classified content to external parties.
- In scenarios involving multiple accounts and devices working together, it's crucial to strictly isolate data from different users, roles, and projects to prevent data sharing, information interference, and content confusion. This is a common problem. To prevent this, we adhere to a core design philosophy: one device for one task; one device responsible for only one job; ensuring the device becomes increasingly specialized and allows for in-depth data accumulation.

8.4 Compliance Output Control Requirements

The training agent establishes a compliant output mindset, ensuring that all output content is free from illegal remarks, inappropriate content, false information, infringing content, and non-compliant language. Externally output documents, plans, and Q&A content must undergo rigorous compliance verification, while internally output content must guarantee professionalism, rigor, and security, comprehensively mitigating office compliance risks and information

security risks.

9. Social tool connection and IM, backend collaborative solution

Note: This chapter focuses on the differences between the crayfish social tools and the backend tools; it does not represent the complete operating logic of our STA-100. The core of our STA-100 is a lightweight AI agent that helps users with daily office work, private data storage, and on-demand data retrieval. Its front-end IM social tools and back-end functions are similar.

If you use our STA-100 connected to a monitor to communicate with OpenClaw using our OpenClaw backend, the core tools of our OpenClaw backend can serve as your inference tool for communication and inference between the user and OpenClaw. We do not recommend opening multiple web pages simultaneously through our STA-100 device. As a lightweight intelligent agent, opening too many web pages will overload the network and significantly slow down the operation. Communicating and inferring with OpenClaw solely through the OpenClaw web backend is the most efficient method.

The following work rules are based on your hardware setup, which includes more than one laptop; communication and emphasis will differ in this case.

To further enhance the practicality, ease of use, scenario coverage, and 24/7 service capabilities of the crayfish intelligent agent, this chapter standardizes the connection methods, port activation processes, and permission adaptation rules for intelligent agent social tools. It clarifies the logic for combining and using IM instant messaging tools with the backend system, subdivides usage scenarios, and provides precise usage suggestions. This creates a complete office loop that integrates deep backend configuration, lightweight online IM interaction, and the implementation of work content, maximizing the value of intelligent agents in office work.

9.1 Core Connection Methods of Social Tools

The OpenClaw intelligent agent fully supports rapid integration and stable mounting with mainstream social and instant messaging tools, enabling multi-port online monitoring, 24/7 message response, instant task execution, and convenient mobile office work . Its core connection logic is standardized and unified, following a four-step process: **backend configuration and binding, IM port activation, permission adaptation and activation, and scenario adaptation and debugging**. No complex deployment or professional technical operation is required; ordinary users can quickly complete the mounting and configuration, achieving the normalized, stable, and standardized use of the intelligent agent in social tools.

Users can independently complete a full set of settings through the backend system, including device binding, account authorization, port opening, scene adaptation, and permission management. This allows the intelligent agent to be stably integrated into their frequently used IM social tools, enabling them to be activated anytime, anywhere, interact with commands instantly, and complete tasks immediately. This completely eliminates the limitations of a single backend login operation and significantly improves the convenience of daily office work. It also supports independent binding of a single account, binding of multiple accounts for teams, and simultaneous online access for multiple devices, perfectly adapting to both individual and team collaborative work scenarios.

9.2 Standalone Use Cases and Recommendations for IM Tools

The IM instant messaging port boasts core advantages **such as lightweight, high frequency, fast response, fragmented processing, and mobile adaptation** . It focuses on daily fragmented office work, temporary and sudden needs, and instant Q&A interaction. It is the most frequently used and most convenient core port for users, emphasizing efficiency, time-saving, and on-demand accessibility.

- **Handling routine lightweight needs** : Quick copywriting

modification, short sentence creation, speech optimization, basic Q&A, simple information retrieval, short sentence polishing, basic data organization, format fine-tuning and other temporary lightweight tasks can be executed quickly by waking up the intelligent agent directly through the IM port, without logging into the backend or performing complicated operations, saving office time efficiently.

- **Real-time interactive communication** : For interactive needs such as instant consultation on work-related questions, streamlining work ideas, correcting copywriting logic, optimizing expression, polishing initial drafts of solutions, and rapid problem analysis, we leverage the real-time, second-level response capabilities of IM to achieve immersive, continuous dialogue and interaction, and efficiently streamline work logic.
- **Mobile 24/7 Office** : Whether working remotely, from home, or without a computer, users can access intelligent agents anytime through social IM tools, breaking down limitations of devices, locations, and times, and enabling them to work anytime, anywhere and handle ad-hoc work needs.

Usage Recommendations: **The IM port is positioned as a lightweight interactive port, intended only for temporary, fragmented, low-load, and routine tasks. It is not recommended to perform high-load, high-precision, and long-term operations such as complex model training, large-scale document processing, massive data aggregation, in-depth rule configuration, or long-term parameter tuning on the IM port, as this may consume port computing power, reduce response speed, and lead to decreased output stability.**

9.3 Independent Use Scenarios and Recommendations for Backend Systems

The backend system focuses on **professionalism, depth, configurability, long-term sustainability, high load capacity, and the ability to accumulate** core capabilities. It serves as the core operating platform for the intelligent agent's underlying training,

parameter debugging, capability iteration, data accumulation, rule configuration, and risk management . It undertakes all deep, core, and long-term operations and is the core foundation for the stable operation and continuous evolution of the intelligent agent.

- **Core underlying parameter configuration** : All core underlying configuration operations, such as Soul/Agent/Memory/Skill four core modules role calibration, underlying working rule updates, model default selection, operation permission settings, invalid memory cleanup, skill library optimization, etc., must be completed uniformly in the background to ensure accurate parameter modification and traceable retention.
- **Deep and long-term training iteration** : Long-term training work such as batch updating of the dedicated training language library, importing industry professional knowledge base, reviewing and optimizing problems at each stage, iterating and upgrading the capabilities of the intelligent agent, and adjusting output standards are all carried out in a unified system through the backend and archived to form a complete training record.
- **For batch complex and high-load tasks** such as batch processing of large numbers of documents, creation of in-depth content of tens of thousands of words, complex business logic deduction, planning of long-term work plans, and multi-dimensional data summary and analysis, which have high load, high precision, and high difficulty requirements, background operation is given priority. By relying on the computing power advantage of the background, the output quality, stability and integrity are guaranteed.
- **Full data accumulation and review** : All training records, work logs, problem logs, iteration records, model usage data, and deviation records of the intelligent agent are automatically saved and archived in the background, supporting query, review, and export at any time, providing complete data support for long-term capability optimization, work review, and standard accumulation.

9.4 Core Mechanism for IM+ Backend Collaboration

(Recommended Standard Solution)

The combination of lightweight IM interaction and deep backend empowerment is the official, standardized, most efficient, and most suitable implementation solution of Xiaolongxia Intelligent Agent. It perfectly realizes a two-way closed-loop linkage of **lightweight and efficient front-end office work and long-term deep backend empowerment**, taking into account both daily office efficiency and long-term capability iteration.

9.5 Robots form groups and cooperate to assist humans in their work.

In our practical application, we established a company group using Lark/Feishu, which included not only company employees but also a group of bots, each assigned a specific job role. As the bots' training deepened and their communication with humans improved, the group's workflow became increasingly seamless, significantly enhancing the organization's internal productivity. For a small business, deploying multiple intelligent agents can drastically reduce labor costs and increase productivity.

9.6 Complete SOP process for linking social media tools (WeChat, Telegram, Feishu, etc.)

We will provide separate adaptation processes for Lark and other similar services. Based on our experience, WeChat and Telegram are relatively easy to bind, and our product also has direct plugins; this is the best IM solution for users.

Professional productivity tools like Lark and WhatsApp can be complex for the average person, generally requiring specialized engineers for configuration and troubleshooting. We provide a professional Lark compatibility guide to help users utilize OpenCLAW-Lark as a productivity tool for small internal teams. Lark's interface offers a more professional presentation for everyday work. Currently, for our STA-100, which serves as a lightweight personal assistant, we recommend using WeChat for

communication; it's perfectly adequate.

The standardized process of mounting, binding, and debugging a social IM agent is readily available, allowing even beginners to easily implement and complete the configuration with zero barriers.

- **Step 1: Enable backend permissions** : The administrator enables IM port permissions, configures the device whitelist, defines the scope of usable functions, and initializes the basic port parameters in the backend.
- **Step 2: Account Authorization and Binding** : The user initiates social account authorization in the background, completes account association and device binding, and confirms that the binding status is normal and there are no missing permissions.
- **Step 3: Scene adaptation and debugging** : Log in to the IM port, test functions such as regular Q&A, lightweight creation, and basic query, and verify that the output style, character design, and logic are consistent with the backend.
- **Step 4: Port rule solidification** : Lock the lightweight usage permissions of the IM port and disable the backend advanced configuration, batch operation, and parameter modification entry points to avoid accidental frontend operations.
- **Step 5: Daily Inspection and Acceptance** : Check the online status, response stability, and output consistency of multiple ports daily. If disconnection or abnormal response occurs, rebind and debug in a timely manner.

9.7 Frequently Asked Questions (FAQ) (New)

- **Q1: How to deal with slow or laggy IM response?**

A: Prioritize terminating current high-load tasks, clearing temporary conversation memories, restarting lightweight tasks, migrating high-load tasks to the background, and restoring port smoothness.

- **Q2: Why is the output style of IM inconsistent with the backend?**

A: The backend uniformly calibrates Soul and Agent parameters,

synchronously refreshes the memory and rules of both ends, retests and compares, solidifies the unified output standard, and regularly synchronizes parameters in both directions.

- **Q3: Why does memory crosstalk occur when multiple accounts are bound?**

A: The backend enables a data isolation mechanism to clean up crosstalk, solidify the scenario rules for each account separately, and strictly distinguish between personal/team and different project data.

- **Q4: Why do intelligent agents frequently misunderstand?**

A: We collect scenarios with misunderstandings, optimize dedicated instruction keywords, supplement scenario constraints, input them into the training library for specific solidification, and conduct retesting to thoroughly resolve the problem.